

Electrolytic conduction

- **Electrolytic conduction** is the process by which an electric current is carried through a substance due to the movement of ions. This typically occurs in **electrolyte solutions** (e.g., salts dissolved in water) or **molten ionic compounds**.
- **Electrolytes** are substances that dissociate into **positive (cations)** and **negative (anions)** ions in solution or when molten. These free ions are responsible for conducting electricity.
- Examples:
 - **NaCl (sodium chloride)** $\rightarrow$ Na^+ and Cl^- in water.
 - **HCl (hydrochloric acid)** $\rightarrow$ H^+ and Cl^- in water.

Conductivity

- **Conductivity depends on:**
 - Concentration of ions
 - Type of electrolyte (strong vs weak)
 - Temperature (higher temp → more mobility)
- **No conduction occurs in solid ionic compounds** because ions are fixed in a lattice.
- **Molten salts and aqueous solutions** can conduct electricity due to mobile ions.

Introduction to Ionic Migration

- **Electric Field Effect:** Ions migrate toward oppositely charged electrodes when an electric current is passed through an electrolyte solution.
- **Unequal Speeds:** Cations and anions move at different velocities (u_+ and u_-) depending on their size, charge, and hydration.
- **Current Distribution:** Total current flowing through the solution is the sum of currents carried by individual ions ($I = I_+ + I_-$).

Types of Electrolytes

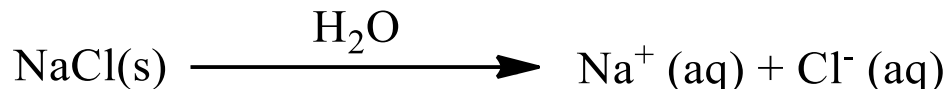
Type	Behavior in Water	Example
Strong	Fully dissociates into ions	NaCl, HCl
Weak	Partially dissociates	CH ₃ COOH
Non-electrolyte	Does not dissociate	Glucose

Electrolytic conduction-mechanism

- **Dissociation:**

The electrolyte dissociates into its constituent ions in solution (or upon melting).

Example:



- **Application of Electric Field:**

When electrodes are connected to a power source and placed in the electrolyte, an electric field is established.

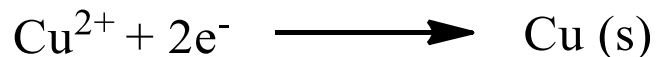
- **Movement of Ions:**

- **Cations (positive ions)** move towards the **cathode (negative electrode)**.
- **Anions (negative ions)** move towards the **anode (positive electrode)**.

- **Electrode Reactions:**

Redox reactions occur at the electrodes:

- At the **cathode** (reduction): Cations gain electrons.



- At the **anode** (oxidation): Anions lose electrons.



- **Current Flow:**

The movement of ions completes the electric circuit, allowing current to flow.

Transport Number and Kohlrausch's Law

What is Transport Number?

- A transport number, also called a transference number, represents the fraction of the total electric current carried by a specific ion in an electrolytic solution. It's a measure of how effectively an ion contributes to the overall conductivity of the electrolyte.
- Transport numbers are crucial in understanding the ionic conductivity of solutions and the role of individual ions in electrolysis.

Transport Number

$$t_+ = \frac{\text{Current carried by cation}}{\text{Total current passed through the solution}}$$

$$t_- = \frac{\text{Current carried by anion}}{\text{Total current passed through the solution}}$$

$$t_+ = \frac{I_+}{I_+ + I_-} = \frac{u_+}{u_+ + u_-} \qquad t_- = \frac{I_-}{I_+ + I_-} = \frac{u_-}{u_+ + u_-}$$

$$\text{So, } t_+ + t_- = \frac{I_+}{I_+ + I_-} + \frac{I_-}{I_+ + I_-} = \frac{I_+ + I_-}{I_+ + I_-} = 1$$

Key Constraint: The sum of transport numbers always equals one ($t_+ + t_- = 1$).

Factors Affecting Transport Number

- **Nature of Ions:** Highly hydrated ions move slower, decreasing their transport number.
- **Concentration:** Changes in concentration alter inter-ionic attractions, affecting ion speeds differently.
- **Temperature:** High temperatures reduce hydration, bringing the values of t_+ and t_- closer to 0.5.
- **Nature of Co-ions:** The transport number of a specific ion changes if its counter-ion is swapped.

Methods to Determine Transport Number

- - Hittorf's Method (concentration change near electrodes)
- - Moving Boundary Method (sharp ion boundary movement)

Kohlrausch's Law

- Full Name: Kohlrausch's Law of Independent Migration of Ions
- Statement: At **infinite dilution**, where inter-ionic forces become negligible, each ion migrates completely independently of its co-ion.
- The limiting molar conductivity (Λ_m^0) of an electrolyte is equal to the sum of the limiting ionic conductivities of its individual cations and anions.
- Mathematical Equation: $\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$
(Where ν_+ and ν_- are the stoichiometric coefficients of cations and anions, and λ^0 represents limiting ionic conductivity).

Relation Between Transport Number & Kohlrausch's Law

- At infinite dilution, limiting ionic conductivities are directly proportional to ionic mobilities (u).
- Therefore, the limiting transport number can be expressed as:

$$t_+^0 = \frac{\lambda_+^0}{\Lambda_m^0} \quad \text{and} \quad t_-^0 = \frac{\lambda_-^0}{\Lambda_m^0}$$

Where,

$$\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0 \quad (\Lambda_m^0 = \text{limiting molar conductivity})$$

λ_+^0 = molar ionic conductivity of cation;

λ_-^0 = molar ionic conductivity of anion

ν_+ and ν_- are the stoichiometric coefficients of cations and anions,

λ^0 represents limiting ionic conductivity

- This link allows us to calculate individual ionic conductivities directly from measured transport numbers.

Applications of Kohlrausch's Law

- **Weak Electrolytes:** Calculates the limiting molar conductivity (Λ_m^0) of weak electrolytes (like acetic acid) which cannot be determined by extrapolation.
- **Degree of Dissociation (α):** Determines how much a weak electrolyte splits into ions:
 - $\alpha = \frac{\Lambda_m}{\Lambda_m^0}$
- **Dissociation Constant (K_a):** Aids in calculating the equilibrium constant for weak acids and bases.
- **Solubility Products:** Determines the solubility (S) of sparingly soluble salts like AgCl.

Conclusion

- Transport numbers measure the relative current workload shared by individual ions.
- Kohlrausch's Law provides a reliable framework to calculate electrochemical properties at infinite dilution.
- Together, these concepts are fundamental for studying conductometric titrations, batteries, and industrial electrolysis.